

Role of altered *rpoB* alleles in *Bacillus subtilis* sporulation and spore resistance to heat, hydrogen peroxide, formaldehyde, and glutaraldehyde

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Abstract Mutations in the RNA polymerase β -subunit gene *rpoB* causing resistance to rifampicin (Rif^R) in *Bacillus subtilis* were previously shown to lead to alterations in the expression of a number of global phenotypes known to be under transcriptional control. To better understand the influence of *rpoB* mutations on sporulation and spore resistance to heat and chemicals, cells and spores of the wild-type and twelve distinct congenic Rif^R mutant strains of *B. subtilis* were tested. Different levels of glucose catabolite repression during sporulation and spore resistance to heat and chemicals were observed in the Rif^R mutants, indicating the important role played by the RNA polymerase β -subunit, not only in the catalytic aspect of transcription, but also in the initiation of sporulation and in the spore resistance properties of *B. subtilis*.

Keywords *Bacillus subtilis* · Sporulation · Spore resistance · Heat · Chemicals

Introduction

Bacillus subtilis is considered a ubiquitous microorganism that can be isolated from almost every niche in nature (Nicholson et al. 2000, 2002; Horneck et al. 2010). Dormant spores exhibit astounding longevity up to thousands of years (Gest and Mandelstam 1987; Cano and Borucki 1995; Nicholson et al. 2000; Vreeland et al. 2000). In the dormant state, damage to crucial spore components (i.e., DNA) accumulate and can only be repaired after the initiation of germination, the first step in the return of spores to vegetative growth (Setlow and Setlow 1996; Setlow 2003, 2006). It has been shown that the developmental cycle of *B. subtilis*, consisting of postexponential-phase gene expression, sporulation, dormancy, and germination, is controlled in large part at the transcriptional level (reviewed in Paidhungat and Setlow 2002; Piggot and Losick 2002; Setlow 2003).

The ability of microorganisms to survive and proliferate in their habitats depends on a variety of molecular mechanisms that adjust gene expression patterns in response to changing environmental conditions (Bandow et al. 2002; Nicholson and Maughan 2002; Gaynor et al. 2005; Saint-Ruf and Matic 2006; Perkins and Nicholson 2008; Kang and Park 2010). DNA-dependent RNA polymerase (RNAP) is a central macromolecular machine controlling the flow of information from genotype to phenotype, and insights into global transcriptional regulation can be gained by studying mutational perturbations in the enzyme (Ingam and Furneaux 2000; Maughan et al. 2004; Perkins and Nicholson 2008). RNAP is composed of an essential catalytic core enzyme ($\alpha_2\beta\beta'\omega$) and one of several alternative sigma (σ) factors (reviewed in Helmann and Moran Jr. 2002). Because the RNAP holoenzyme contacts every promoter in the genome, single amino acid substitutions in

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critical portions of the enzyme can lead to global changes in gene expression, hence in physiology and metabolism (Maughan et al. 2004; Perkins and Nicholson 2008).

Rifampicin (Rif) inhibits transcription initiation by binding to the β -subunit of bacterial RNAP and blocking the mRNA exit channel (Boor et al. 1995; Wehrli et al. 1968; Wehrli and Staehelin 1971; Wehrli 1977; Campbell et al. 2001). Mutations in the nucleotide sequence within the *rpoB* gene encoding the β -subunit can cause rifampicin resistance (Rif^R). Rif^R mutations have been isolated from numerous bacterial species and mapped to four small areas within the *rpoB* gene called Clusters I, II, III, and N (reviewed in Campbell et al. 2001; Severinov et al. 1993). To date, all of the Rif^R mutations isolated in *B. subtilis* have been found to be single nucleotide substitutions resulting in specific amino acid changes within the N-cluster and Cluster I (Nicholson and Maughan 2002; Maughan et al. 2004; Perkins and Nicholson 2008). Previous reports demonstrated that certain Rif^R mutations altering amino acids Q469, H482 and S487 in Cluster I of *B. subtilis rpoB* exhibit alterations in fundamental physiological processes such as growth rate, competence for transformation, sporulation, and germination (Rothstein et al. 1976; Maughan et al. 2004; Perkins and Nicholson 2008).

To date, however, limited information is available regarding the impact of different *rpoB* alleles on spore formation and spore resistance properties. Therefore in this work, we have examined the effects of mutations in the *B. subtilis rpoB* gene on the regulation of sporulation initiation and spore resistance to obtain a better understanding of the mechanisms involved in multi-factorial nature of spore resistance (as reviewed in Setlow 2006).

Materials and methods

Bacterial strains

Strains used in this work are listed in Table 1, and all are congenic with the wild-type strain WN547 (MH5636). In this work, strain WN547 is referred to as the wild-type strain with respect to RNAP, and all twelve congenic mutant strains are referred to by the amino acid substitution carried in their RpoB β -subunit.

Measurement of glucose catabolite repression during sporulation

To determine glucose catabolite repression during sporulation, the wild-type and the Rif^R *rpoB* mutant strains were cultured in liquid Schaeffer Sporulation Medium (SSM) (Schaeffer et al. 1965) with and without a supplement of 1 % glucose under vigorous aeration (250 rpm) at 37 °C as described previously (Maughan et al. 2006). After 24 h, the colony-forming units (CFUs) before and after heat treatment (80 °C, 10 min) were determined to calculate the sporulation frequency (i.e., the ratio of heat-resistant spores (S) to total viable cells (V)) of the wild-type and the Rif^R *rpoB* mutant strains as described in detail previously (Nicholson and Setlow 1990; Maughan et al. 2006).

Spore preparation and purification

Spores of all strains were obtained by cultivation under vigorous aeration in SSM for 5 days at 37 °C until a sporulation frequency of >95 % had been reached, as judged by phase-contrast microscopy. Where appropriate, chloram-

Table 1 *B. subtilis* strains used in this study

Strain	Genotype and/or phenotype	Source (reference)
WN547 (MH5636)	<i>trpC2 pheA1 cat 10His-rpoC</i> ; Cm ^R	Perkins and Nicholson (2008)
WN758	<i>trpC2 pheA1 rpoB</i> -Q469R <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN759	<i>trpC2 pheA1 rpoB</i> -H482R <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN760	<i>trpC2 pheA1 rpoB</i> -H482Y <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN761	<i>trpC2 pheA1 rpoB</i> -S487L <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN999	<i>trpC2 pheA1 rpoB</i> -S487F <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1000	<i>trpC2 pheA1 rpoB</i> -S487Y <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1002	<i>trpC2 pheA1 rpoB</i> -H482D <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1004	<i>trpC2 pheA1 rpoB</i> -V135F <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1007	<i>trpC2 pheA1 rpoB</i> -Q469K <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1009	<i>trpC2 pheA1 rpoB</i> -H482P <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1011	<i>trpC2 pheA1 rpoB</i> -Q469L <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	Perkins and Nicholson (2008)
WN1191 ^a	<i>trpC2 pheA1 rpoB</i> -A478D <i>cat 10His-rpoC</i> ; Cm ^R Rif ^R	This study

^a The Rif^R *rpoB* allele A478D arose spontaneously from vegetative cells of the *B. subtilis* strain 168 (Moeller et al. unpublished data) was transferred into strain WN547 and verified by PCR amplification and nucleotide sequencing as described previously (15, 21), resulting in strain WN1191

phenicol (Cm) (5 µg/ml) and/or Rif (50 µg/ml) were added to the SSM. Sporulated cultures were harvested by centrifugation (10,000×g, 20 min, 4 °C) and treated with MgSO₄ (2.5 µg/ml), lysozyme (200 µg/ml) and DNase I (2 µg/ml) for 30 min at 37 °C in order to destroy the residual vegetative cells. The enzymes were inactivated by heating for 10 min at 80 °C. After repeated centrifugation and washing in distilled water, the purified spores (about 10¹⁰ spores/ml) were stored in aqueous suspension at 4 °C. Spore preparations consisted of single spores with no detectable clumps and were free (>99 %) of growing cells, germinated spores and cell debris, as seen in the phase-contrast microscope (Nicholson and Setlow 1990; Moeller et al. 2007, 2011).

Measurement of spore resistance

To assay spore resistance to chemicals and heat, spore samples consisting of 10⁷ spores were exposed to heat (90 °C wet heat or 100 °C dry heat) and chemicals (final concentrations of 2 % formaldehyde, 2 % glutaraldehyde, 5 % hydrogen peroxide in water) as previously described (Popham et al. 1995; Riesenman and Nicholson 2000; Tennen et al. 2000; Paredes-Sabja et al. 2008). Spore recovery and viability assays were performed as described previously (Riesenman and Nicholson 2000; Tennen et al. 2000; Moeller et al. 2007, 2011).

Determination of spore viability and statistical analyses

The surviving fraction of *B. subtilis* spores was determined from the ratio N/N_0 , with N the number of colony-forming units (CFU) of the treated sample and N_0 that of the non-treated controls. Spore inactivation curves, representing dose-effect correlations, were obtained as described (Moeller et al. 2011). Data are reported as LD₉₀-values, the time (in h or min) required to kill 90 % of the initial spore population (Riesenman and Nicholson 2000; Tennen et al. 2000; Moeller et al. 2007, 2011). Data are expressed as averages ± standard deviations. All experiments measuring spore resistance were repeated at least three times, and data were compared statistically using Student's *t*-test. Values were analyzed in multigroup pairwise combinations, and differences with *P* values of ≤0.05 were considered statistically significant (Riesenman and Nicholson 2000; Tennen et al. 2000; Moeller et al. 2007, 2011).

Results

Glucose catabolite repression during spore formation in different *rpoB* alleles

In *B. subtilis*, sporulation initiation is triggered when nutrient deprivation activates a protein phosphorylation cascade

(known as the “phosphorelay”) that ultimately phosphorylates the Spo0A protein, the master transcriptional activator of early sporulation genes (reviewed in Perego and Hoch, 2003). Glucose-mediated catabolite repression prevents initiation of sporulation by reducing the expression of genes whose products are needed in the phosphorelay system (Shafikhani et al. 2003a, b). Therefore, a standard assay to measure glucose repression of sporulation initiation is simply to compare the sporulation efficiency of a strain cultivated in SSM vs. SSM containing 1 % (v/v) glucose (SSM + G). To test the effect of glucose catabolite repression of sporulation initiation, we grew wild-type parent strain WN547 and each of the 12 isogenic Rif^R strains at 37 °C in liquid SSM and SSM + G, and the efficiency of sporulation was quantified by determination of the ratio of heat-resistant spores (S) to total viable cells (V) from 24-h cultures (Fig. 1). By 24 h all cultures had grown to equivalent cell concentrations, averaging between 2.91×10^8 and 8.25×10^8 cfu/ml in SSM and SSM + G, respectively. No culture deviated from the average by more than a factor of 2.6, indicating that the mutant *rpoB* alleles exerted no dramatic effect on growth or 24-h cell yield. The wild-type parental strain WN547 sporulated efficiently at 37 °C in SSM with ($S/V = 0.35 \pm 0.12$) and sporulation was severely repressed in SSM + G ($S/V = 0.0045 \pm 0.0019$) (Fig. 1), indicating the high efficiency of glucose acting as sporulation repressor (Inaoka and Ochi 2007; Maughan et al. 2006; Schaeffer et al. 1965). Isogenic strains carrying the Q469L, H482P, or S487Y mutations sporulated in SSM at higher efficiencies than did the parental strain, whereas strains carrying the V135F, Q469K, Q469R, A478D, H482R, H482Y mutations sporulated at a significantly lower frequency in SSM than the wild-type strain (Fig. 1). No significant differences in the sporulation efficiency were obtained in the strains carrying H482D, S487F and S487L mutations (Fig. 1). In contrast, when analyzing sporulation efficiency in SSM + G, the strains carrying the Q469L, S487F, S487L, S487Y mutations sporulated at higher efficiencies than the wild-type strain, whereas the strain carrying the H482Y mutation consistently sporulated at a significantly lower frequency in SSM + G (Fig. 1). No significant differences in the sporulation efficiency in SSM + G were obtained for the strains carrying the V135F, Q469K, A478D, H482D, H482R mutations compared to the wild-type *rpoB* strain (Fig. 1). The levels of glucose catabolite repression during sporulation, as determined in the ratio of the sporulation frequency in SSM with the sporulation frequency in SSM-G, were ranging from 203 to 1.1, with only the H482Y mutation causing a significantly higher glucose catabolite repression rate compared with the wild-type *rpoB* strain (Fig. 2). Strains carrying the mutations Q469K, Q469L, A478D, H482P, H428R, S487F, S487L, and S487Y were all significantly more glucose

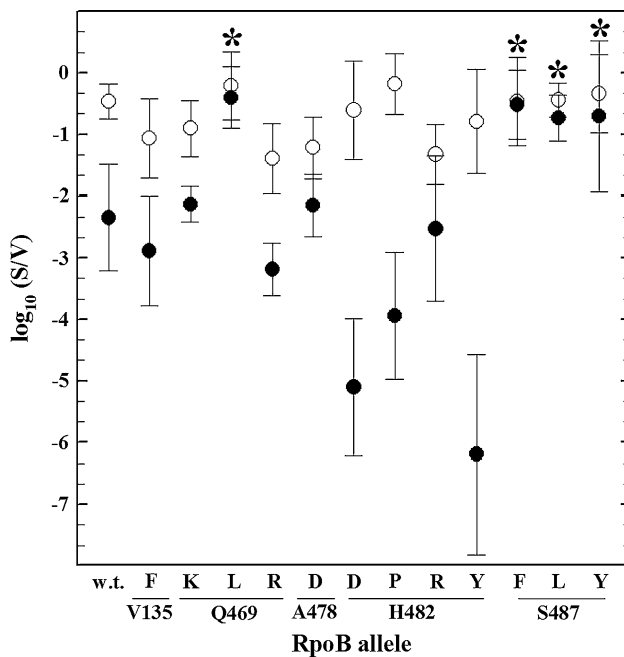


Fig. 1 Glucose catabolite repression of sporulation in the wild-type and Rifampicin-resistant (Rif^R) *rpoB* mutant strains of *B. subtilis*. Sporulation frequency (S/V) in the absence (white circles) and presence (black circles) of 1 % glucose are presented. Data are expressed as averages $\pm$ standard deviations ($n = 3$). Asterisks denote non-significant differences ($P > 0.05$)

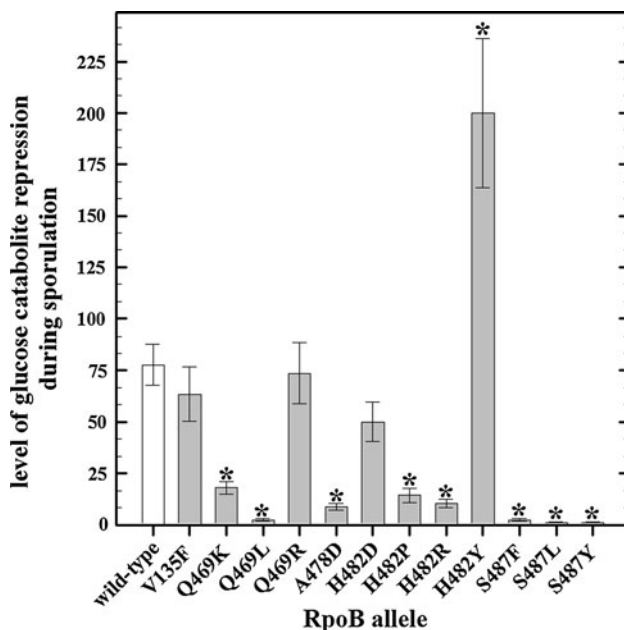


Fig. 2 Levels of glucose catabolite repression during sporulation in the wild-type (white bar) and Rifampicin-resistant (Rif^R) *rpoB* mutant strains (gray bars) of *B. subtilis*, as determined by the ratio of the sporulation frequency of the two different sporulation approaches (with and without a glucose supplement). Data are expressed as averages and standard deviations ($n = 3$). Asterisks indicate glucose catabolite repression levels that were significantly different ($P \leq 0.05$) compared to the wild-type *rpoB* strain

catabolite repression resistant than wild-type, whereas strains carrying the V135F, Q469R, or H482D mutations exhibited wild-type levels of catabolite repression of sporulation (Fig. 2).

Spore resistance to dry and wet heat

In order to determine the impact of altered *rpoB* alleles on the spore resistance to heat, spores were prepared from the wild-type and the isogenic Rif^R strains were subjected to 100 °C dry heat (Fig. 3a) and 90 °C wet heat (Fig. 3b). Semi-logarithmic plots of spore survival versus exposure time yielded the time in h (dry heat) or min (wet heat) required to kill 90 % of the spore population (LD₉₀), and the average LD₉₀ values from at least three separate experiments were compared (Fig. 3; Table 2). The LD₉₀ value of the dry- and wet-heat-treated wild-type *rpoB* spores (with a *His*-tagged *rpoC*) was 4.6 ± 0.2 h and 17.4 ± 3.5 min, respectively, which is in good agreement with previous observations of *B. subtilis* spores with a normal RNAP (without a *His*-tagged *rpoC*) (reviewed in Nicholson et al. 2000; Setlow 2006). The LD₉₀ value of the dry- and wet-heat-treated wild-type *rpoB* spores (with a *His*-tagged *rpoC*) was 4.6 ± 0.2 h and 17.4 ± 3.5 min, respectively, which is in good agreement with previous observations of *B. subtilis* spores with a normal RNAP (without a *His*-tagged *rpoC*) (reviewed in Nicholson et al. 2000; Setlow 2006). Spores of strains carrying a V135F, Q469R, H482Y, and S487Y *rpoB* mutation were significantly more dry-heat-resistant than the wild-type, and only the spores of the Q469K *rpoB* mutant strain were significantly more dry-heat-sensitive (Fig. 3a). In all the other tested *rpoB* mutant spores, no significant differences in the dry-heat resistance were observed (Fig. 3a). When comparing the LD₉₀ values from the wet-heat spore treatments of the *rpoB* mutants, spores carrying the V135F, Q469L, Q469R, and H482P mutations were significantly more resistant to the wild-type spores, whereas spores carrying the Q469K, A478D, H482D, S487L, or S487Y mutations were significantly more sensitive to wet heat than the wild-type spores (Fig. 3b). No significant differences in wet-heat resistance were determined for spores carrying the H482R, H482Y, or S487F mutations (Fig. 3b). Interestingly, when taken together, only spores carrying a V135F were significantly more resistant to both wet- and dry-heat exposure (Fig. 3). Of the 12 mutants tested, only spores carrying the H482R or S487F mutation showed wild-type spore resistance levels to both dry and wet heat (Fig. 3).

Spore resistance to chemicals

In addition to heat resistance, we measured the spore resistance to a variety of chemicals, namely 2 % formaldehyde, 2 % glutaraldehyde, and 5 % H₂O₂ (Fig. 4), with these treatments chosen for comparison with literature data (as reviewed in Nicholson et al. 2000; Setlow 2006). The LD₉₀ value of formaldehyde- and glutaraldehyde-treated wild-type *rpoB* spores (with a *His*-tagged *rpoC*) was 13.6 ± 2.4 and 23.6 ± 1.8 min, respectively, and again those survival char-

Fig. 3 Resistance of spores from wild-type (black circle) and Rifampicin-resistant (Rif^R) *rpoB* mutant strains (white circles) of *B. subtilis* to 100 °C dry heat (a) and 90 °C wet heat (b). LD₉₀ values are expressed as averages ± standard deviations (*n* = 3). One asterisk indicates LD₉₀ values that were significantly lower than wild-type spores, and two asterisks denote LD₉₀ values that were significantly higher than wild-type spores (*P* ≤ 0.05). Note the difference in the y-axes between a and b

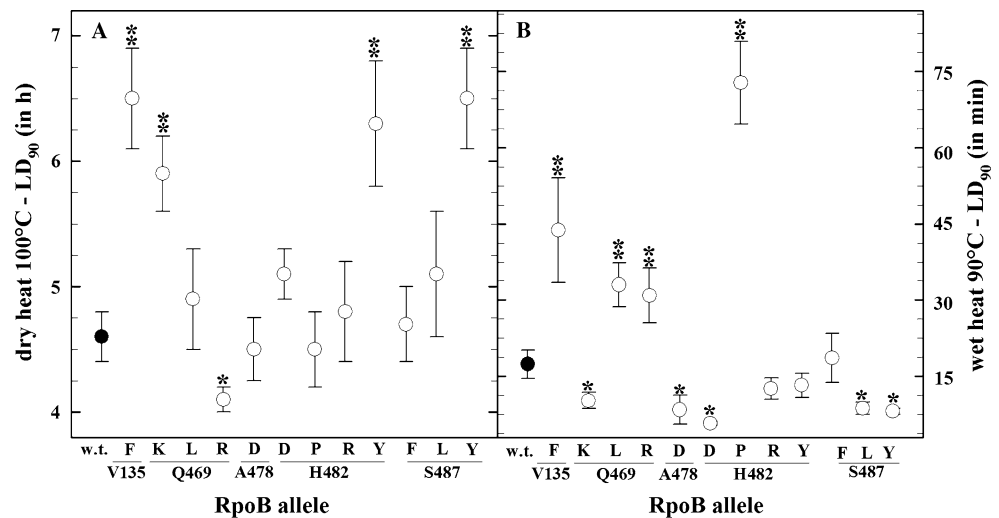


Table 2 Survival characteristics of wild-type and *rpoB* mutant spores of *B. subtilis*

RpoB genotype	Dry heat (100 °C) ^a	Wet heat (90 °C) ^b	Formaldehyde (2 %) ^b	Glutaraldehyde (2 %) ^b	Hydrogen peroxide (5 %) ^b
Wild-type	4.6 ± 0.2	17.4 ± 3.5	13.6 ± 2.4	23.6 ± 1.8	43.7 ± 1.9
V135F	6.5 ± 0.4 ^I	43.8 ± 9.7 ^I	6.0 ± 1.2 ^R	10.4 ± 2.0 ^R	32.9 ± 2.6 ^R
Q469K	5.9 ± 0.3 ^I	10.2 ± 1.6 ^R	1.4 ± 0.3 ^R	17.9 ± 2.8 ^R	29.7 ± 2.9 ^R
Q469L	4.9 ± 0.4 ^U	32.9 ± 4.3 ^I	15.9 ± 3.6 ^U	12.6 ± 2.6 ^R	29.1 ± 2.3 ^R
Q469R	3.9 ± 0.1 ^R	30.8 ± 5.4 ^I	3.6 ± 0.7 ^R	9.0 ± 3.2 ^R	40.5 ± 2.8 ^U
A478D	4.5 ± 0.3 ^U	8.4 ± 2.8 ^R	4.8 ± 0.6 ^R	16.7 ± 4.4 ^R	28.7 ± 3.9 ^R
H482D	5.1 ± 0.4 ^U	5.7 ± 0.4 ^R	13.4 ± 2.2 ^U	21.7 ± 3.4 ^U	28.4 ± 2.6 ^R
H482P	4.5 ± 0.3 ^U	72.6 ± 8.1 ^I	8.5 ± 3.1 ^R	30.1 ± 2.1 ^I	46.5 ± 1.3 ^U
H482R	4.8 ± 0.4 ^U	12.8 ± 2.1 ^R	24.9 ± 2.7 ^I	28.7 ± 2.4 ^I	40.1 ± 3.8 ^U
H482Y	6.3 ± 0.5 ^I	13.3 ± 2.7 ^R	14.9 ± 2.4 ^U	16.5 ± 2.9 ^R	33.1 ± 1.9 ^R
S487F	4.7 ± 0.3 ^U	18.7 ± 4.8 ^U	20.2 ± 4.1 ^I	11.9 ± 2.4 ^U	34.2 ± 1.4 ^R
S487L	5.1 ± 0.5 ^U	8.8 ± 1.2 ^R	12.6 ± 2.7 ^U	24.2 ± 3.8 ^U	51.4 ± 3.1 ^I
S487Y	6.5 ± 0.4 ^I	8.2 ± 0.6 ^R	26.9 ± 3.3 ^I	32.7 ± 2.6 ^I	46.3 ± 3.4 ^U

The characters (R, U, I) indicate reduced (R), increased (I) or unaffected (U) spore resistance compared to the wild-type *rpoB* strain

Data are averages and standard deviations (*n* = 3)

^a LD₉₀ values (i.e., decimal reduction) in h (for the dry-heat treatment)

^b LD₉₀ values (i.e., decimal reduction) in min (for wet heat, formaldehyde, glutaraldehyde, or hydrogen peroxide treatment)

acteristics were in good agreement with previous data obtained from *B. subtilis* spores with a normal RNAP (without a *His*-tagged *rpoC*) (Loshon et al. 1999; Melly et al. 2002; Paidhungat et al. 2000; Nicholson et al. 2000; Setlow 2006). Spores of strains carrying the H482R, S487F, or S487Y mutations were significantly more resistant to formaldehyde, whereas spores of the V135F, Q469K, Q469R, A478D, or H482P mutations were more formaldehyde-sensitive than wild-type spores (Fig. 4a). No significant differences in formaldehyde resistance were determined for spores carrying the Q469L, H482D, or S487L mutations (Fig. 4a; Table 2). Determination of spore glutaraldehyde resistance (Fig. 4B) revealed a spectrum of resistance similar to that of formaldehyde resistance (Fig. 4a) for spores carrying the

V135F, Q469R, H482D, H482R, S487L, or S487Y mutations (Fig. 4b). Differences were noted in that: spores carrying the mutations Q469K and A478D were found to be significantly more sensitive to formaldehyde but not glutaraldehyde; spores carrying the Q469L and H482Y mutations were significantly more sensitive to glutaraldehyde but not formaldehyde; spores carrying the H482P allele were significantly more resistant to glutaraldehyde but not formaldehyde; and spores with the S487F mutation were more sensitive to glutaraldehyde but more resistant to formaldehyde than wild-type spores (compare Fig. 4a, b).

Analysis of resistance to 5 % hydrogen peroxide showed that only spores carrying a S487L mutation were significantly more resistant to H₂O₂ than wild-type spores

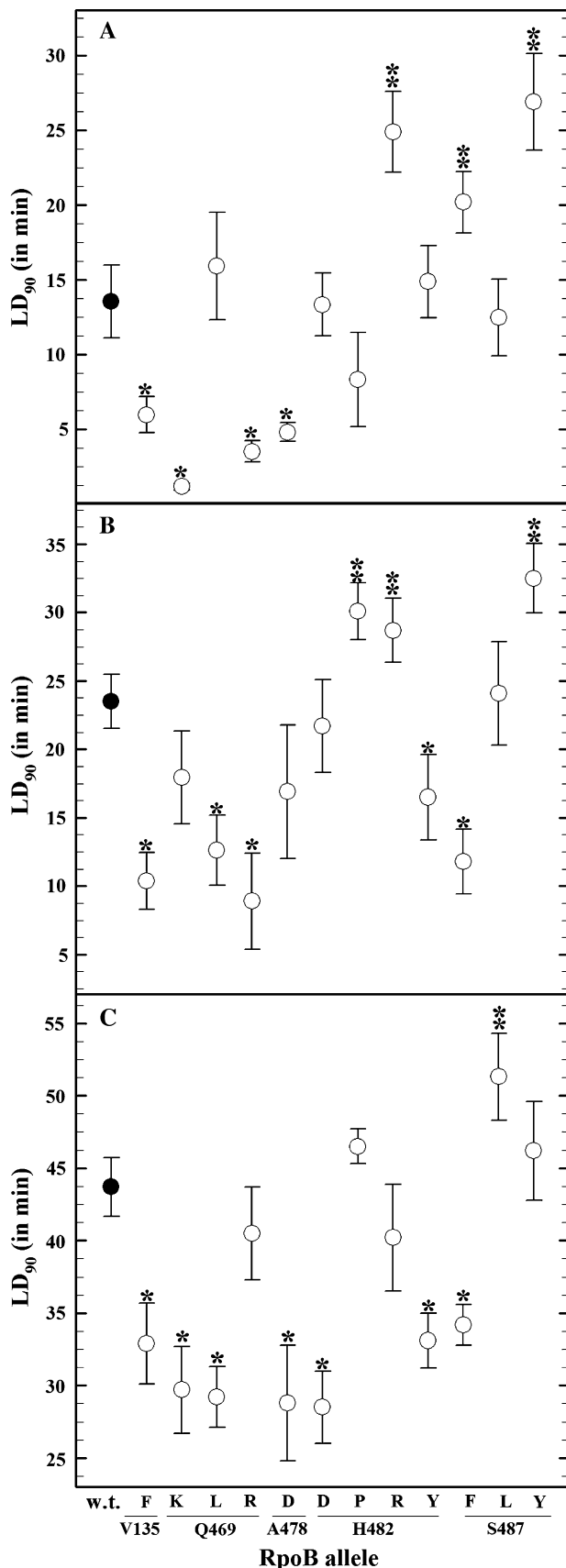


Fig. 4 Resistance of spores from wild-type (black circle) and Rifampicin-resistant (Rif^R) *rpoB* mutant strains (white circles) of *B. subtilis* to 2 % formaldehyde (a), 2 % glutaraldehyde (b), and 5 % hydrogen peroxide (c). LD₉₀ values are expressed as averages $\pm$ standard deviations ($n = 3$) as described in the text. One asterisk indicates LD₉₀ values that were significantly lower than wild-type spores, and two asterisks denote LD₉₀ values that were significantly higher than wild-type spores ($P \leq 0.05$). Note the difference in the y-axes between a, b and c

(Fig. 4c). Spores of strains carrying the Q469R, H482P, H482R, or S487Y behaved essentially like wild-type spores (Fig. 4c), while spores carrying a V135F, Q469K, Q469L, A478D, H482D, H482Y, or S487F exhibited significant lower H₂O₂ resistance levels than the wild-type *rpoB* spores (Fig. 4c). Of all the *rpoB* mutants tested, only the strain carrying the V135F mutation was significantly more sensitive to all chemical treatments than were wild-type spores (Fig. 4; Table 2).

Discussion

It has been shown that the developmental cycle of *B. subtilis*, consisting of postexponential-phase gene expression, sporulation, dormancy, and germination, is controlled in large part at the transcriptional level (reviewed in Paidhungat and Setlow 2002; Piggot and Losick 2002). Here, we present data indicating that alterations in closely linked amino acids residing within *rpoB* N-cluster (V135F) and cluster I (Q469, A478, H482, and S487) not only confer Rif^R, but also exert a range of effects on the catabolite repression of sporulation initiation and on spore resistance properties. In addition to the Rif^R phenotype, each amino acid change in *rpoB* resulted in its own unique spectrum of catabolite repression and spore resistance (Fig. 1; Table 2).

Glucose catabolite repression

Cells enter the pathway to sporulate in response to conditions of nutrient limitation, which results in the formation of the predivisional cell (Wang et al. 2006). The initiation of sporulation is a complex process by which environmental signals are sensed by the cell and transduced via a phosphorelay system to the master transcriptional regulator Spo0A (reviewed in Perego and Hoch 2002). Glucose is one of the most effective repressors of sporulation initiation (Schaeffer 1969), and its effect has been traced to down-regulation of transcription of genes encoding components of the phosphorelay system (*spo0F*, *kinA*) and to *spo0A* itself (Shafikhani et al. 2003a, b). A catabolite-resistant sporulation (Crs) phenotype was previously found to result from a missense mutation (*crsA447*; leading to P290F) in the *sigA* (*rpoD*) gene encoding the

major vegetative RNAP sigma-A (σ^A) factor (Kawamura et al. 1985). Thus, the direct involvement of the vegetative σ^A subunit of RNAP in glucose catabolite repression of sporulation has been demonstrated. In our experiments, strains carrying the Q469L, S487F, S487L, S487Y mutations in *rpoB* also exhibited a strong Crs phenotype (Fig. 1), thus also implicating the β -subunit of RNAP in glucose catabolite repression of sporulation initiation. It is interesting to note that in a separate but relevant study, Nanamiya et al. (2000) isolated a mutation in *sigA* (E314K) that caused sporulation initiation to be temperature sensitive. A second-site repressor mutation restoring sporulation at high temperature was isolated and subsequently mapped also to the *rpoB* gene (A863G), strongly implicating direct interaction between the β and σ subunits of RNAP in both activation and catabolite repression of sporulation initiation (Nanamiya et al. 2000; Helmann and Moran Jr. 2002).

Spore resistance properties

Spores are extremely resistant to being killed by heat and a variety of cytotoxic chemicals, and spore resistance properties can be traced to unique structures of the spore, such as a highly cross-linked protein coat, a modified peptidoglycan spore cortex, a low core water content and abundant intracellular spore core constituents such as the calcium chelate of dipicolinic acid (Ca-DPA) and α/β -type small, acid-soluble spore proteins (α/β -type SASP), the latter two of which protect spore DNA (Hackett and Setlow 1988; Popham et al. 1995; Driks 1999; Nicholson et al. 2000; Paidhungat et al. 2000; Setlow et al. 2000; Lee et al. 2008). The genes responsible for these structural proteins and spore-specific enzymes are transcriptionally controlled in either the developing forespore or mother cell during differentiation (for extensive reviews, see Piggot and Losick 2002; Driks 1999, 2002; Paidhungat and Setlow 2002), thus alterations in RNAP affecting promoter binding and selectivity might be predicted to result in altered spore properties. In most (but not all) cases, *Rif^R* mutations in *rpoB* resulted in lowering of spore resistance to various chemical treatments, but complete concordance was only seen in the V135F mutation, which significantly lowered spore resistance to formaldehyde, glutaraldehyde, and H_2O_2 (Fig. 4).

Although specific *rpoB* mutations appeared to increase or decrease the chemical, wet-heat, or dry-heat resistance of spores (Figs. 3, 4), no clear-cut pattern could be discerned; it thus appears that each *Rif^R* mutation exerted its own specific spectrum of effects on each property tested, summarized below.

V135F

The V135F strain showed no differences in the glucose catabolite repression during spore formation (Fig. 1). Inter-

estingly, V135F spores were more resistant to heat, but more sensitive to the tested chemicals than wild-type *rpoB* spores (Figs. 3, 4).

Q469

We tested three different Q469 alleles (K, L, R) on their catabolite repression and spore resistance. Surprisingly, the strain carrying a Q469L mutation was resistant to catabolite repression of sporulation initiation, while the Q469K and Q469R alleles were not (Fig. 2). In regard to their spore heat resistance, only Q469K spores were more resistant to dry heat than wild-type spores, whereas the Q469L and Q469R alleles were significantly more resistant to wet heat than wild-type spores (Fig. 3). None of the Q469 alleles was more resistant to the three tested chemicals than the wild-type (Fig. 4); in fact, spores of the Q469K mutant were most sensitive to formaldehyde of all strains tested (Fig. 4a).

A478D

Sporulation of the A478D strain was substantially more glucose repression-resistant than the wild-type strain (Fig. 2). A478D spores showed the same level of resistance to dry heat than the wild-type, but a weaker level in spore resistance to wet heat and the three tested chemicals (Table 2).

H482

At codon H482, we were able to compare four different amino acid substitutions (D, P, R, and Y), all previously isolated as spontaneous mutations in both vegetative cells and spores (Nicholson and Maughan 2002; Perkins et al. 2008). Only the H482R mutant was found to be resistant to glucose catabolite repression (Fig. 2). Only spores of the H482Y strain were more dry-heat resistant than wild-type spores, while the H482D and H482P strains showed significantly lower or higher wet-heat resistance, respectively (Fig. 3). Regarding spore resistance to chemicals, the H482 mutants exhibited a variety of patterns (Fig. 4). Interestingly, spores of the H482R were more resistant than wild-type spores to formaldehyde and glutaraldehyde, but not to hydrogen peroxide, wet or dry heat.

S487

With codon 487 of *rpoB*, we had the opportunity to compare the physiological effects of replacing the polar amino acid S with either a non-polar (L) or an aromatic (F, Y) amino acid. Considering the rather dramatic chemical differences, the amino acid changes would be predicted to confer, all three mutants exhibited glucose catabolite repression resistance (Fig. 1), whereas numerous allele-specific

differences in spore resistance to heat and chemicals were obtained (Figs. 3, 4; Table 2).

Conclusions

Our results lend support to the hypothesis that the β -subunit of RNAP plays a much wider regulatory role in the spore formation under glucose-limiting and -abundant conditions as well as in the spore resistance to heat and cytotoxic chemicals than has previously been appreciated. Recently, Keijser et al. (2007) identified several key transcriptional events during the first 2 h of spore germination, showing direct correlations between transcriptional and cellular activity. It seems apparent that a well-developed regulatory mechanism exists for the integration of key physiological events during germination and outgrowth. They suggested that there are checkpoint processes interacting on DNA integrity, metabolic status, the initiation and prolongation of spore germination toward vegetative growth (Wang et al. 2006; Keijser et al. 2007). Previous reports by Maughan et al. (Maughan et al. 2004) and others (e.g., Nanamiya et al. 2000; Perkins and Nicholson 2008) showed that Rif^R *rpoB* mutations definitively effect the regulation of developmental processes during sporulation and germination by both locus-specific and allele-specific effects. However, the results reported here clearly establish the importance of the RNAP in the initiation of sporulation as well as in the spore resistance to heat and chemicals. At present, the exact mechanism(s) involved in regulatory role of RNAP is obscure, but it is certainly amenable to experimental elucidation. We are directing current experiments toward a definition of how the different *rpoB* alleles affect in vivo expression of the complete *B. subtilis* transcriptome during sporulation and spore germination (with and without prior exposure to damaging agents or physical stressors). We predict that such studies will lead to the identification of key developmental genes whose expression is altered in *rpoB* mutants and which are thus dependent upon the regulatory function of the RNAP β -subunit.

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